



Complex Engineering Problems Analysis

Process Manual Section 1.5 Compliance, Washington Accord Attributes & Engineering Synthesis

Project: AI-Powered Smart Campus Resource Management System	Faculty Coordinator / Mentor: Dr. Rashmi Mothkur (ASAC CSE)
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1. Compliance with Process Manual Section 1.5 & ABET Criterion 5

Engineering design projects inherently address *Complex Engineering Problems* as defined by ABET Criterion 5, the Washington Accord (Graduate Attributes WA1–WA7), and NBA Program Outcomes (PO1–PO12). In strict conformance with Section 1.5 of the Alliance University ASAC Process Manual, the table below maps each mandated attribute directly to the engineering implementation of the **AI-Powered Smart Campus Resource Management System**:

Complex Problem Attribute	Process Manual Criteria	Implementation in Smart Campus Resource Management System
Deep Technical Knowledge Required (WA1)	Cannot be solved by simple textbook formulas; requires multi-disciplinary engineering synthesis.	Synthesizes machine learning (LSTM demand forecasting), non-linear constraint satisfaction algorithms (CSP room scheduling), IoT streaming protocols (MQTT/WebSockets), 3D WebGL graphics, and thermodynamic chiller energy balance equations.
Open-Ended, Multiple Solutions (WA2)	No single correct answer; engineering trade-offs exist between cost, performance, and reliability.	Navigated engineering trade-offs between 3D polygonal mesh fidelity (LOD) vs. client browser frame-rate (60 FPS), heuristic room allocation speed vs. global theoretical optimum, and peak chiller setback aggressive dispatch vs. occupant comfort (24.0 ± 1.0 °C).
Involves Conflicting Constraints (WA2/P2)	Lightweight vs. strong, low-cost vs. durable, fast vs. energy efficient.	Maximizing classroom seat fill-rate while minimizing student travel between distant campus blocks; aggressive commercial grid peak shaving vs. maintaining academic thermal comfort; rich visual 3D asset detail vs. strict web binary size limits (<15 MB).
Not Fully Defined in Advance (WA4/P4)	Requirements evolve as the team discovers technical, operational, or customer constraints.	Initial project focus was limited to room scheduling; field operational audits revealed 28.4% ghost bookings and severe BESCO maximum demand surcharges, prompting expansion to include automated space intelligence and solar-BESS microgrid balancing.
Requires Information Gathering (WA3/P3)	Literature review, vendor data, international standards (IEEE, ISO), user feedback.	Synthesized 15 peer-reviewed IEEE/Elsevier research publications (2024–2026), BESCO commercial tariff rate cards, Alliance University GIS spatial blueprints, timetable ERP schemas, and semi-structured interviews with facilities electrical engineers.
Non-Standard Solution Path (WA4/P4)	Cannot be solved by routine methods; requires creativity, iteration, judgment.	Constructed a novel closed-loop cyber-physical coupling: real-time occupant spatial density dynamically drives localized chiller setback commands and battery energy storage dispatch in a zero-install browser canvas without proprietary BMS hardware.
Involves Real-World Constraints (WA5/P5)	Budget limits, safety regulations, supply chain issues, environmental impact.	Zero-dollar commercial software budget (100% open-source stack); non-intrusive student privacy regulations (strictly anonymous headcount data, zero biometric facial recognition); IEEE IoT wireless standards and ASHRAE 90.1 energy baselines.
Requires Teamwork & Coordination (WA6/P6)	Different sub-systems must interface correctly; complex integration challenges arise.	Coordinated across 4 distinct specialized subsystems: Central Dashboard & 3D Spatial Engine (Likith S), Energy Analytics & Microgrid Routing (Vinodkumar), IoT Telemetry & Sensor Simulation (Nikhil kumar V), and Timetable Optimization & Space Intelligence (Kalyan kumar T).

2. Curricular Requirements & Phased Deliverables (Section 1.6)

As established in Section 1.6 of the ASAC Process Manual, the Design Project spans two formal academic semesters: **Design Project – I (5CS1990 / E1CSA313) [Semester V]** establishes requirements elicitation, literature synthesis, mathematical modeling, system architecture, and a functional simulation prototype with empirical validation (100% Completed). **Design Project – II (6CS1991 / E1CSA314) [Semester VI]** will deploy physical IoT sensor hardware across campus buildings, integrate direct RESTful APIs with the Alliance University ERP system, and validate multi-seasonal energy performance.